Tyndall National Institute

Utilising Dynamic Motor Control Index to identify Age-Related differences in Neuromuscular Control

Final Internship Report

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Introduction

1.1 Overview

Muscle Synergy (MS) analysis has been used extensively in the neuroscience, sport science and rehabilitation fields to gain greater insight into motor control (Hug, 2011). The Central Nervous System (CNS) recruits the muscles needed for movement in groups known is muscle synergies. This optimises movement by allowing the muscles and joints to operate as a coordinated unit to accomplish a movement goal (Abd et al., 2021). These muscle groupings are not fixed synergies for the range of one's life. A multitude of reasons can cause these synergies to shift and regroup, combine or divide. An example of this is post muscle injury. The CNS regroups the muscles in the synergies to give the injured muscle a less weight-bearing role(Hodges & Tucker, 2011; Lund et al., 2011). Although this new grouping of synergies may be less optimal in the long term, short-term use allows quicker healing of the injured muscle (Schache et al., 2010). The leading MS analysis method, where muscle synergies are extracted using non-negative factorisation or principal component analysis etc, has shown differences in the number of muscle synergy present before and after an injury(Avrillon et al., 2020). Furthermore, it has been seen that often the muscle synergies will not return to their previous most optimal state after rehabilitation for said injury(Avrillon et al., 2020).

It is well known that older people are more prone to trips, falls, and general poor balance (Häkkinen et al., 1996; Pijnappels et al., 2008; *Role of Exercise on Sarcopenia in the Elderly - European Journal of Physical and Rehabilitation Medicine 2013 February;49(1):131-43 - Minerva Medica - Journals*, n.d.). While age-related declines in gait and balance control have been documented (Era & Heikkinen, 1985; Springer et al., 2006), the same cannot be said for muscle synergies. Numerous studies explored the age-related modifications to muscle function/coordination through the use of sEMG sensors monitoring muscle activity during walking. When compared to a younger sample, studies suggest that there is no significant difference in the muscle synergies of that of a young person and that of an old person(Monaco et al., 2010). Even tasks that are more complex such as walking upstairs could not differentiate the neuromuscular control of a young population and an old population based on the number of muscle synergies extracted (Baggen et al., 2020). Although extracting muscle synergies from sEMG activity using dimensionality reduction techniques has been considered a gold standard for observing injury-related differences in muscle synergies, the same cannot be said regarding age-related differences in muscle synergies. Clinical evaluation of elderly people's muscle function is largely carried out through functional assessments such as studying gait and physical energy (Soubra et al., 2019). Although useful, research suggests neuromuscular control declines first, and functional impairments follow (Clark et al., 2013; Dingwell et al., 2017). Therefore, implementing reliable neuromuscular analysis into these evaluations may allow us to identify these impairments before they even begin to affect physical function. Rehabilitation and preventative measures can then be taken to improve muscular function before the issue becomes too serious.

This study utilises a novel analysis method, called the Dynamic Motor Control Index (DMCI), to explore its ability to identify age-related differences in muscle synergies. It was recently used in 2021 to explore its ability as a method of marking neuromotor impairments in older adults in a way that does not rely on the number of muscle synergies (Collimore et al., 2021). However, the study was restricted in terms of exercise used and is the one of the only studies of its type. While the study (Collimore et al., 2021) showed promising results, the average DMCI score of the young adult population and the old adult population overlapped significantly (100+-10 compared to 96.4+-10).

Although Collimore's study was successful at distinguishing the age groups based on the DMCI index, this study aims to reduce the overlap of DMCI scores by using more complex exercises which may better distinguish age-related differences.

Ultimately, this study aims to further explore the potential of DMCI and its relation to the difficulty of the exercises.

1.2 Muscle Synergy Analysis

The aim of MS analysis is to reduce the number of dimensions (muscles) of the EMG to a limited set of smaller, more meaningful variables referred to as synergies. This can be done through dimensionality reduction techniques such as non-negative matrix factorisation or principal component analysis (Turpin et al., 2021). The outcome means to provide insight into the neural strategies for movement and functional outcomes of muscle activities. Those with a fewer number of muscle synergies (one or two) have greater co-activation between their muscles and thus, less optimal muscle control (Schmitz et al., 2009). Those with three or four muscle synergies (i.e. less co-contraction) are thought to have more optimal neuromuscular control (da Silva Costa et al., 2020; Hortobágyi & DeVita, 2006).

The EMG activity of m muscles at a given time $e(t)$ is acquired through the linear combination of a number of synergy vectors, which are then scaled in amplitude by an activation coefficient. (Lena H Ting, 2010):

$$e(t) = \sum_{j=1}^{s} c_j(t)\boldsymbol{w}_j + residual$$

Where s is the number of synergies, which must be less or equal to the number of muscles at a given time. j is the synergy index, $c_j(t)$ is the activation coefficient and w is the synergy vectors. This is then rewritten into matrix form:

$$\boldsymbol{E} = \boldsymbol{W} \times \boldsymbol{C} + \boldsymbol{residual}$$

This depiction of the EMG data separates the spatial (W) and temporal (C) and defines their dimensionality as the minimal number of vectors necessary to explain most of the data variance (variance accounted for, VAF)(Turpin et al., 2021).

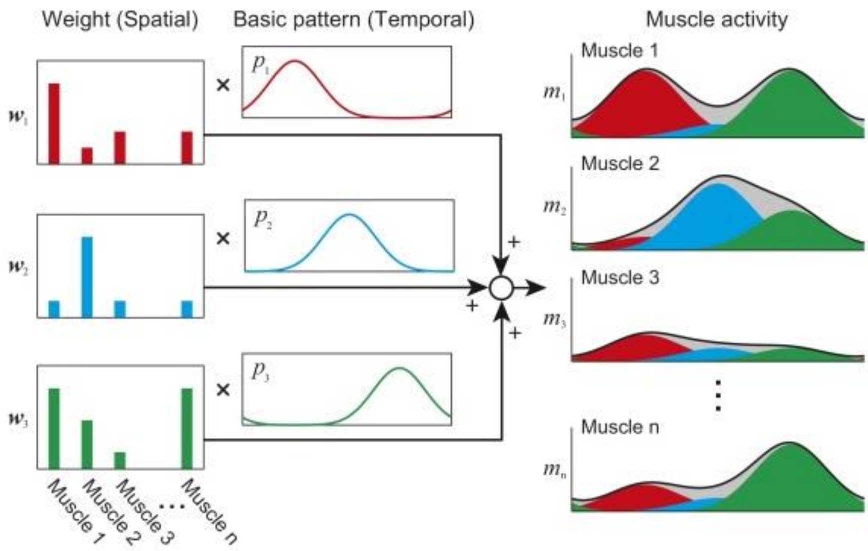


Fig. 1. Muscle Synergy Model: Muscle Activation patterns are produced by basic activation pattern with distribution weights(Aoi & Funato, 2016). The weightings define the direction of the synergy vectors and show how much each muscle increases its activation when a synergy is activated. The temporal structure expresses the activation coefficients as a sequence in time relevant to each other(Turpin et al., 2021).

Muscle synergy analysis consists of three main stages: EMG recording, EMG processing and synergy extraction. Even a brief review of studies executed in this area can confirm the wide variability in the methods preformed in each of these stages. Surface EMG (sEMG) and intramuscular EMG have been used successfully, with preference leaning towards sEMG due to its non-invasive nature(Hug, 2011).The EMG data can be processed in an immense multitude of ways, with many authors reporting that their results are sensitive to their chosen methodology. Choices in normalization methods for the EMG amplitude, band pass filtering threshold frequencies and sampling rate all play a part in producing a signal best suited for MS extraction. In terms of muscle synergy extraction, while non-negative matrix factorisation (NNMF)(Lee & Seung, 1999; Rabbi et al., 2020) and principal component analysis (PCA) (Jackson, 1991; Ranganathan & Krishnan, 2012) are the leading methods, some studies do utilise independent component analysis (Rasool et al., 2016). Full methodology and reasons for choice will be discussed later in this report.

1.2 The Dynamic Motor Control Index (DMCI)

The DMCI analysis method was seen in the context of neuromuscular control when studying the gait in people with cerebral palsy (CP) in 2015 (Steele et al., 2015). It has since been used in two other studies regarding CP patients, children, and adults (Schwartz et al., 2016; Shuman et al., 2019).

The DMCI is a summary metric of muscle co-activation during walking. Rather than computing the number of muscle synergies that correlate to a predefined VAF threshold, the DMCI constrains the number of muscle synergies to one and scales that VAF score to a z-score based on a control group. In the case of investigating age-related differences between a younger group and an older group, the young group is the control group. The higher the DMCI score, the more complex the neuromuscular ability is(Steele et al., 2015). Therefore, it would be assumed, and shown in Collimore's study that younger people have a higher DMC Index than older people because as discussed, older people have more co-activation of their muscles(Collimore et al., 2021). This method is further verified in da Silva Costa's et al. study where the single synergy VAF score differed between younger adults and older adults in a balancing exercise. In comparison, when muscle synergies were analysed, the number of synergies did not differ between the two groups(da Silva Costa et al., 2020).

Single synergy's variance accounted for (VAF) values are computed using the same extraction methods as traditional MS analysis (NNMF, PCA etc.). As mentioned, the single synergy VAF value is then converted to a z-score with the young adult group as the control group:

$$DMCI = 100 + 10\left(\frac{AVG\ VAF_{1synergy-Control} - VAF_{1synergy-Exp}}{SD\ VAF_{1Synergy-Control}}\right)$$

Where $AVG\ VAF_{1Synergy-control}$ is the average of the one synergy VAF of the whole young adult control group, $VAF_{1Synergy-Exp}$ is the one synergy VAF value of each individual in all groups within the study, and $SD\ VAF_{1synergy-control}$is the standard deviation of the single synergy VAF value of the young adult control group (Steele et al., 2015).

Overview of Experimental Procedure

While Collimore's results proved that the DMCI can identify age-related differences in neuromuscular control between an older and younger sample population, there was significant overlap in said results. Younger adults (27$\pm$3years) had an average DMCI score of 100$\pm$10 while 'young-old' adults (73$\pm$3) and 'old-old' adults (78$\pm$2) had an average DMCI score of 96.4$\pm$10.79 and 88.9$\pm$9.67 respectively (Collimore et al., 2021). Statistical analysis showed that this method was capable of observing significant differences between the young adults and the old-old adults. It was not however, able to distinguish between the young-old and old-old age group. It was hypnotised that this inability may be due to the simplicity of the walking task and suggested that other walking tasks that varied in complexity may be the key to identifying age related impairments at an earlier age(Collimore et al., 2021). This is the key take away as we aim to identify age related deficits as young as possible. While collimore's older group's average age was 75 years, our older population's average age is 56 years.

This study aims to extend the scope of this research by utilising the same analysis methods in Collimore's study but, in addition to their simple walking task, more complex exercises such as incline walking, walking up stairs and lateral stepping up stairs will also be analysed. Perhaps, as suggested a more complex task will reduce the overlap seen in the DMCI of the different age groups. If a more complex task can create more distinguishable brackets for the groups' DMCI score with little overlap, this analysis method can be promoted more confidently as a useful means of examining neuromuscular control in adults.

Experimental Parameters

3.1 Study Population

Data from 30 participants was collected. Participants were grouped into two groups: young (18-49 years) and 'old' (50-65 years). They study allowed any gender who had a BMI below 35. The young control group could not have any history of knee injury/disorders in any knee. The older group must have healthy knees, or knees which have sufficiently recovered from past knee trauma and are capable of preforming daily tasks with ease.

	N	Sex	Average Age	Average height	Average Weight
Young Adults	16	9 male 7 female	27.93+-7.5y	171.93+-6.86cm	75.19+-14kg
Older Adults	15	9 male 6 female	56.07+-6.37y	175.24+-8.47cm	78.54+-8.92kg

3.2 Muscles

Included muscles were the primary muscles in the participant's dominant leg. The tensor fasciae latae was removed from the protocol as sEMG sensors was to be in the iliac region which was reported to be uncomfortably close to the suprapubic region. sEMG signal analysis showed large amounts of crosstalk in this signal so the decision to remove the muscle from analysis was made.

Final list of included muscles are: the rectus femoris, vastus lateralis, vastus medialis, biceps femoris caput longus, lateral gastrocnemius, medial gastrocnemius, soleus, semitendinosus, tibialis anterior, peroneus longus, erector spinae, gluteus maximus and the gluteus medius.

E1	☑	Right	Rectus femoris	EMG
E2	☑	Right	Vastus medialis	EMG
E3	☑	Right	Vastus lateralis	EMG
E4	☑	Right	Tibialis anterior	EMG
E5	☑	Right	Biceps femoris caput longus	EMG
E6	☑	Right	Semitendinosus	EMG
E7	☑	Right	Gastrocnemius medialis	EMG
E8	☑	Right	Gastrocnemius lateralis	EMG
E9	☑	Right	Peroneus longus	EMG
E10	☑	Right	Soleus	EMG
E11	☑	Right	Erector spinae ileocostalis	EMG
E12	☑	Right	Tensor fasciae latae	EMG
E13	☑	Right	Gluteus maximus	EMG
E14	☑	Right	Gluteus medius	EMG

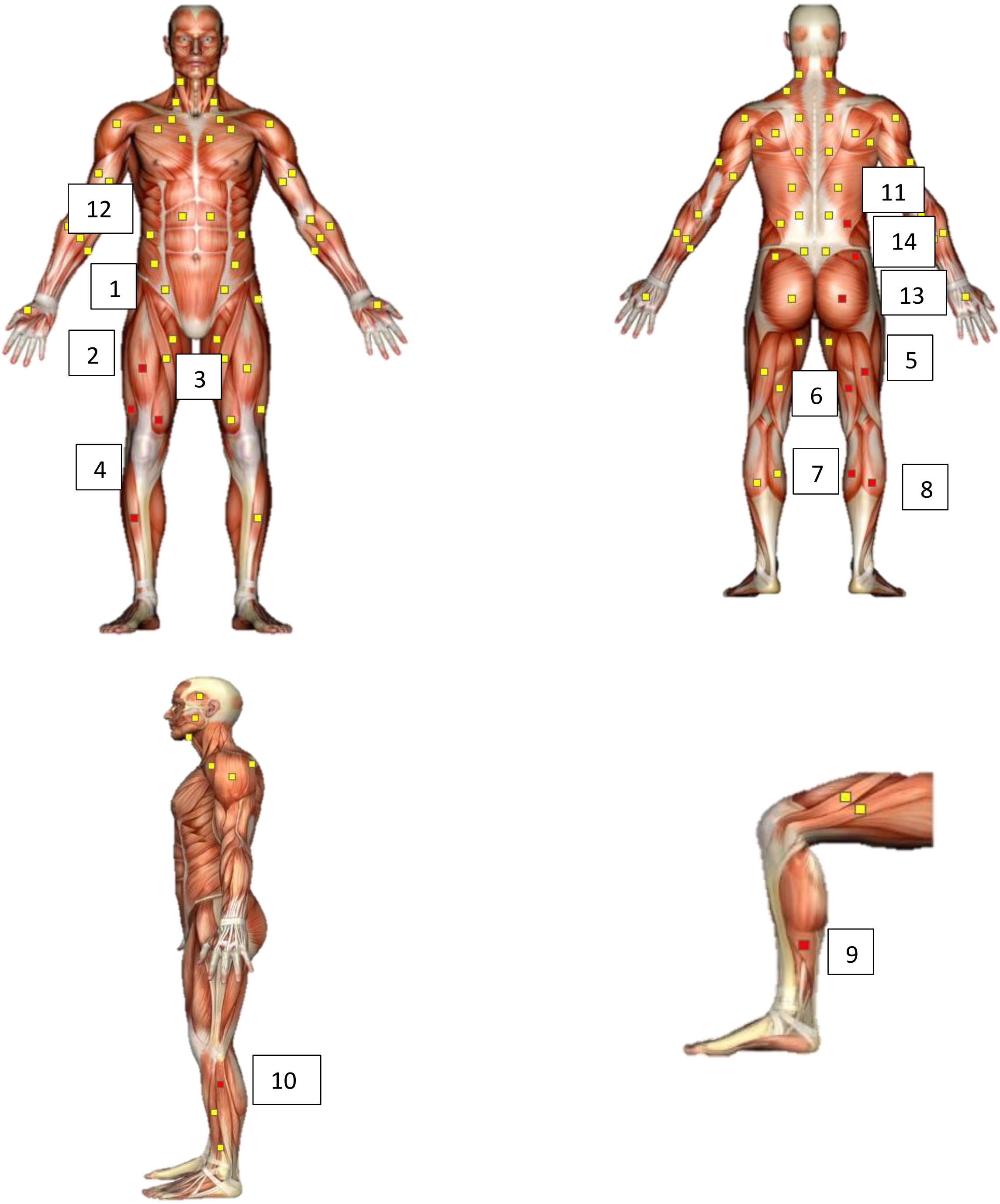


Figure: Experimental Protocol for sEMG sensor placement.

3.3 Sensor Utilisation

Following non-invasive sensors and systems were used in the data collection:

Off the shelf BTS FREEEMG Electromyography Units (CE Class IIa, FDA registered device). The FREEEMG system uses light miniaturized probes with active electrodes for signal acquisition and transmission for EMG assessments. Sensor was attached to the skin of the participants over the muscles of interest using adhesive electrodes. Sensors were placed according to SIEMAN guidelines. A specialised protocol on the BTS FREEEMG software was created correlating each sensor to a specific muscle. These sensor-muscle correlations remained consistent throughout the course of the trial. (see sensor placement in appendix A)

Figure 3. Left: BTS FREEEMG sensor. Right: BTS FREEEMG docking station (unknown, n.d.).

Off the self XSENS MTW IMU (accelerometry) sensors. MTW is a highly accurate, completely wireless, small and lightweight 3D human motion tracker. Sensor collected orientation data only (pitch, roll, yaw). The sensors were placed using elastic straps around the thigh, calf and foot of the participant. (See sensor placement in Appendix B)

Figure 2. Left: Xsens Sensors used in trial. Right: Xsens docking station (Monique Paulich)

3.3 Exercises

As there were 16 sEMG sensors available to use throughout these trials it was decided to collect muscle activity data from the participant’s dominant leg only. This allowed us to collect more muscles (14) rather than only recording 8 muscles on each leg. Participants are asked not to lean on equipment to aid balance unless completely necessary. Three participants in the old group required balance/stability aid during the exercises. (Appendix C)

Any exercise can be practised if requested by the participant.

Flat walking

The treadmill is set to 4km/hr. Data collection begins 10 seconds after the participant starts walking to ensure the treadmill is at the correct speed. 40-50 seconds of data is be collected. The treadmill is then be turned off and participants come to a complete stop with feet around shoulder width apart and legs straight. The researcher notifies participant that data collection has finished and participants may relax. This concludes the first trial.

Incline walking

The treadmill is set to an incline of 5° and a speed of 3.5km/hr. Data collection begins 10 seconds after the participant starts walking to ensure the treadmill is at the correct speed. 40-50 seconds of data is be collected. The treadmill is then be turned off and participants come to a complete stop with feet around shoulder width apart and legs straight. The researcher notifies participant that data collection has finished and participants may relax. This concludes the second trial.

Step Forward

The participant is to begin with both feet close together on the ground. The dominant leg is the leading foot. Data collection commences and participant is advised to start the exercise. The participant steps onto the first step with their leading foot. Their trailing foot follows so both feet are on the same step. This is one rep. Pause for 2 seconds. Participants then repeat the movement onto the second step, where both feet finish on the second step. And again onto the third step. They stay on this third step until the researcher indicates the data collection has been finished. The participant can then come back down in any way they like.

There is be a handrail (the treadmill) beside the steps. Participants can use this if absolutely necessary but are advised not to lean onto it. It is merely a balance aid and should not take away the effort needed from the legs. This is the end of trial 3.

Step Lateral

The participant is to begin with both feet close together on the ground, facing sideways so the stairs are on either their left or right hand side. Their dominate leg should be the inside leg (i.e. closest to the step. Data collection commences and participant is advised to start the exercise. The participant steps sideways onto the first step with their leading foot. Their trailing foot follows so both feet are sideways on the same step. This is one rep. Pause for 2 seconds. Participants then repeat the movement onto the second step, where both feet finish sideways on the second step. And again onto the third step. They stay on this third step until the researcher indicates the data collection has been finished. The participant can then come back down in any way they like.

MS Processing Parameters

4.1 Normalisation

Utilising a maximal voluntary contraction is one of the most powerful and popular normalisation method within EMG processing(Rubega et al., 2021). It allows one to observe the relative activation level of a muscle by expressing the amplitude of the muscle signal during a given exercise as a percentage of a reference EMG value. The reference EMG in this case is the peak/mean value/s of an EMG signal recorded during a participant's isometric maximal voluntary contraction (Meldrum et al., 2009)(i.e. asking the participant to contract a specific muscles as hard as possible).

A secondary aim of this study is to promote a protocol that is able to be used in a variety of settings. Those with muscle impairments and neuromuscular disfunctions (such as osteoarthritis, cerebral palsy, stroke etc) may not be able to voluntarily contract on command. It has been seen that while using this normalisation method, a data value would carry a value greater than 100% of the normalised value(Halaki & Ginn, 2012; Perry et al., 2010). To remain inclusive to all for this protocol, it was elected to normalise the EMG signal to the maximum activation experienced during the exercise trial of which the EMG data was collected from.

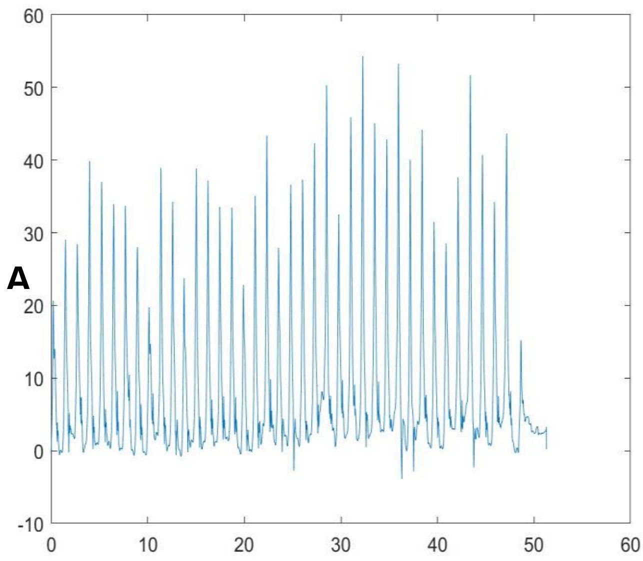
4.2 Filtering

Filtering methods followed those in Collimore's study. The EMG signals were filtered through a 4^{th} order Butterworth filter (high-passed) at 40Hz. It was then detrended (where the sample mean was subtracted from each observation so that they are mean zero), rectified, passed through a 4Hz low pass signal (4th order Butterworth) (Collimore et al., 2021). This filtering was executed in custom MATLAB files made by the author (Laura Burke).

4.3 Time Normalisation

For the two walking exercises, only the middle 30 strides were used for signal processing. Foot gait was recorded through the XSENS sensors and motion data was imported into MATLAB. As the recording frequency is 100Hz, it was oversampled to 1000Hz to match that of the EMG signal frequency. The pitch data was isolated and the first and last five seconds of the recording was removed. The centre maximum peak was located as the beginning of the middle stride. The 30 strides (identified as the 31 peaks) around this centre was located and corresponding EMG data was saved. All data outside of these 30 strides was discarded. The BTS and XSENS devices were synchronised using a hardware trigger.

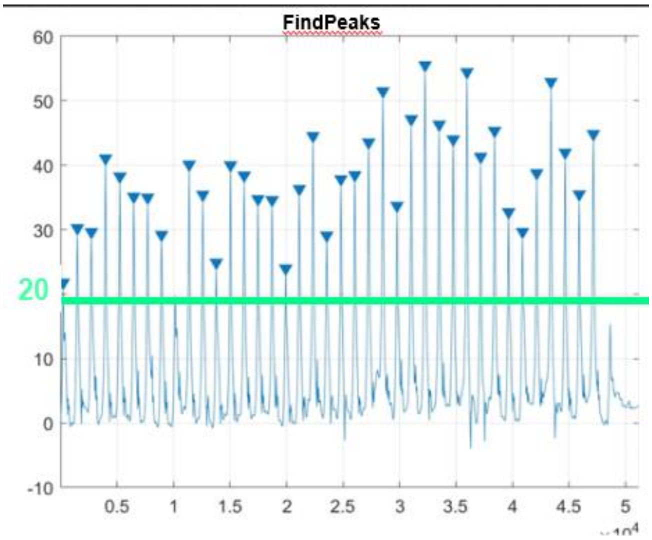
RESAMPLED XSENS



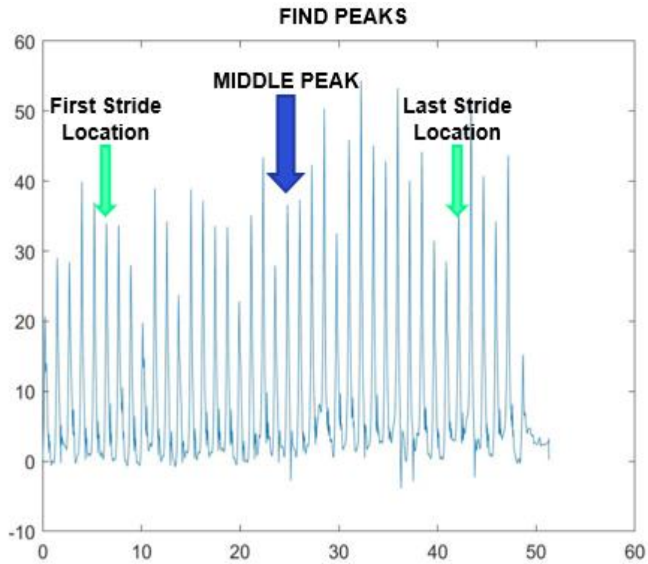
Raw xsens data was resampled to 1000Hz to have the same number of data points as the muscle data.

Each peak represents the beginning of a stride.

Resampled xsens was inverted so to use the findpeak function.

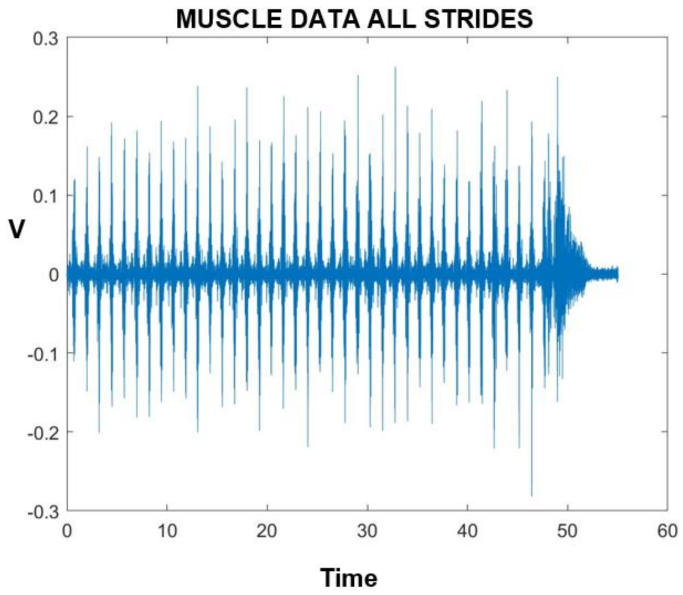


Minimum Peak Amplitude Threshold was set to 20 in this example.



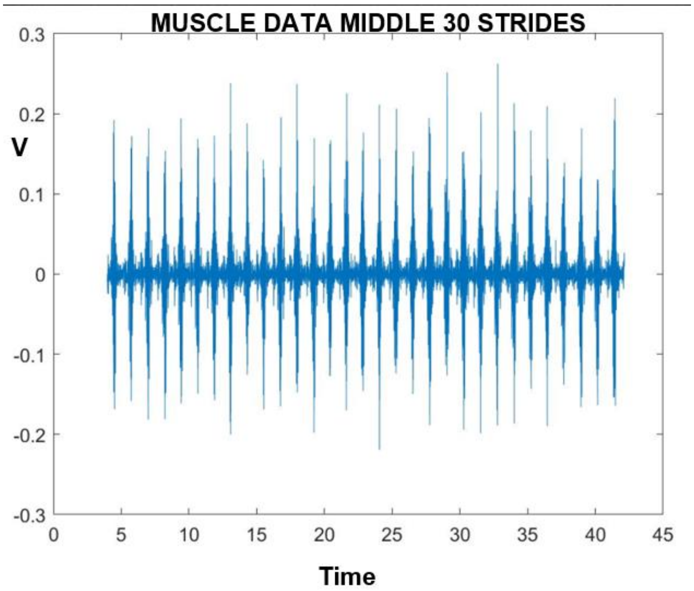
The median is the middle peak.

The first stride location is found 15 peaks to the left of the middle peak. The last of the 30 strides occur 15 peaks to the right of the middle peak.



Muscle data of all strides.

Approximately 55s long.



Muscle data of all the middle 30 stride. Note the discarded data before 5 seconds and after 40 seconds

Muscle Synergy Extraction

This study utilises non-negative matrix factorisation (NMF) to exact synergies from the collected muscle signals. This is the most popular extraction method, being used in over 62% of studies during 1999 to 2018(Rabbi et al., 2020). NMF constrains the synergies to be positive values. Simply put, it is an algorithm that iterates an initial matrix that is non-negative (D) into two matrices W and C (see 'muscle synergy analysis' section above). The algorithm updates these matrices with the multiplicative rules (Rabbi et al., 2020)

$$W_{ik} \leftarrow W_{ik} \frac{(DC^T)_{ik}}{(WCC^T)_{ik}} \text{ and } C_{kj} \leftarrow C_{kj} \frac{(W^T D)_{kj}}{(WW^T C)_{kj}}$$

this avoids the non-increasing trend of the Frobenius norm (the square root of the sum of the squares of the elements of the matrix (Ford, 2015)) of the error matrix D-WC for each iteration.

The variance accounted for (VAF) statistically compared the reconstructed and original muscle patterns(Tresch et al., 2006). It defines the dimensionality of the muscle data in terms of the smallest number of vectors required to explain most of the variability in the data (Kubota et al., 2021). The user predefines a threshold which consistently ranges from 80-95% in the literature(Turpin et al., 2021). VAF was set to a threshold of 95% based on observation of MS extraction results.

Once extracted, the single synergy value was noted for the DMCI calculation, along with the number of muscle synergies present as the corresponding VAF value based on the stated threshold.

Statistical Analysis

Binomial regression tested if the DMCI was significantly predictive of age group after adjusting for the number of muscle synergies. It also tested their interaction. The significance level was set at 0.05.

Multinomial logistic regression used the same predictors to compare the young adult and old adult group. It also tested the predictor's interaction.(Collimore et al., 2021)

Results

Due to timing constraints only preliminary analysis has been completed for the two step exercises. Walking analysis is ongoing.

Step Forward Group	Average DMCI Score
Young Adults (18-49y)	100 +-10
Older Adults (50-64y)	97.94 +- 9.15

Step Lateral Group	Average DMCI Score
Young Adults (18-49y)	100 +-10
Older Adults (50-64y)	98.18 +- 6.15

It appears that despite stairs being classified as a more complex task, the DMCI score is similar to those in Collimore's walking study. However, binary linear regression shows that the DMCI in this case is not significantly able to identify age related differences in DMCI for both steps forward and steps lateral.

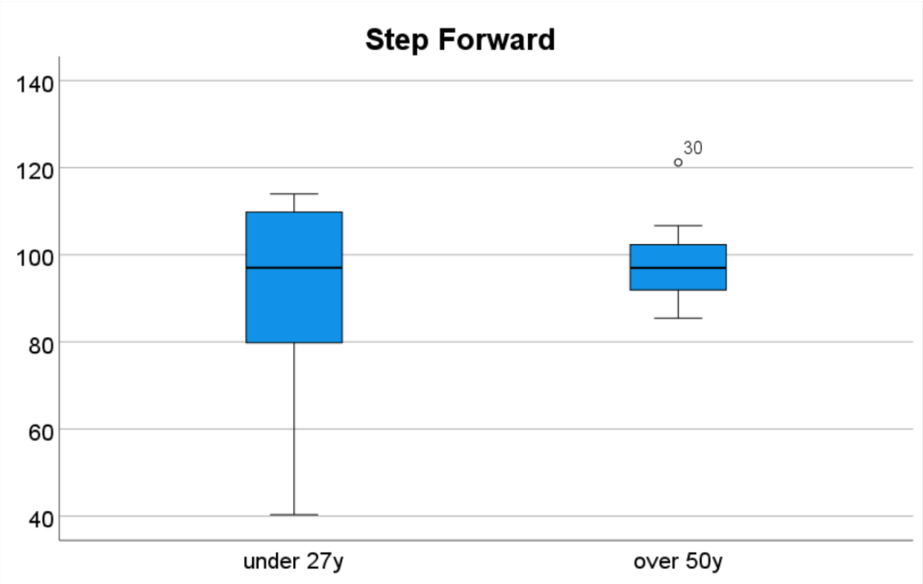
The Regression was unable to accurately predict a participant's age group based on the DMCI or number of muscle synergies. The accuracy was in the range of 60-65% in both instances.

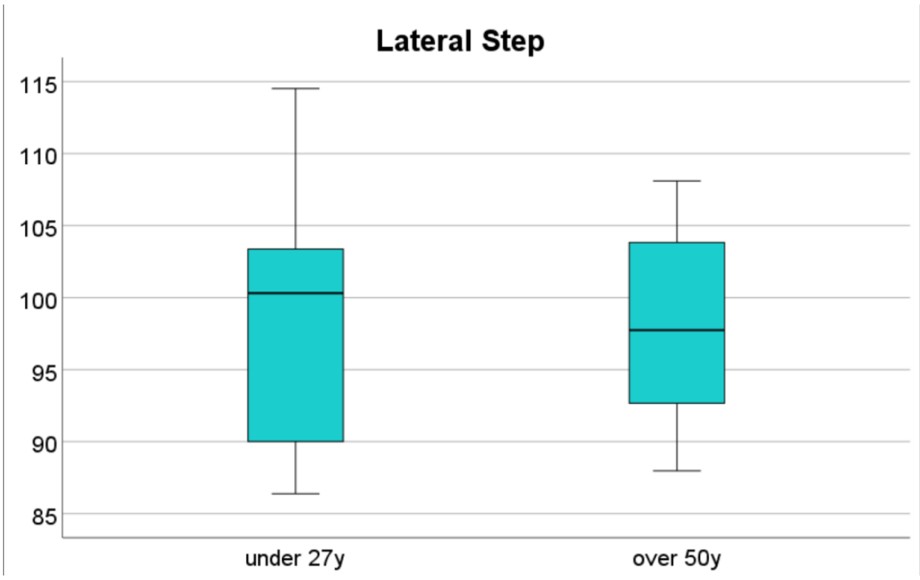
The difference in age between the oldest participant in the young adult group to the youngest participant in the older adult group is 1 year. This is a vast contrast to the 30year difference in Collimore's study. Therefore, the Young adult group was remade to only hold participants between the ages of 18 to 27 years. This group only has 8 participants in it.

Step Forward Group	Average DMCI Score
Young Adults (18-**27**y)	89.9667 +- 28
Older Adults (50-64y)	97.94 +- 9.15

Step Lateral Group	Average DMCI Score
Young Adults (**18-27y**)	99.22 +- 9.4
Older Adults (50-64y)	98.18 +- 6.15

Although the difference between age groups seem smaller, the regression model preferred this set up. It gave a significance level of 0.134 which is still not significant but an improvement from the previous grouping. The accuracy of prediction was also increased to 78%. One obvious issue however is the young adults DMCI for step forward is now lower than the older adult score – this goes against the literature which states the young adults have a higher DMCI score than the older adults (Collimore et al., 2021).





Plotting the results for the under 27y age group to the older adults group clearly shows the hige amount of overlap in the two result groups.

It is clear that despite the more complex exercise, the DMCI failed to identify age related differences in neuromuscular control.

Discussion

As mentioned, these are just preliminary results. I now wish to continue analysis of the step data to check its ability to identify age related differences via the DMCI score. I will preform more regression, and multinominal regression to see what the results produce. I also want to try reprocessing the data with a VAF of 90% (copared to 95%) as that would produce a different number of synergies for the threshold VAF. As the number of muscle synergies is used in the regression model, this may help significance levels.

Data sets will be published along with the findings of this paper.

Conclusion

Although preliminary data analysis does not show promising results for this method, it is still to soon to tell. The method itself is still far too novel with only 4 papers reported and of which inly one in the area of normal muscle function. This method has the potential to promote healthier, more optimal muscle function in older peoples standard of living.

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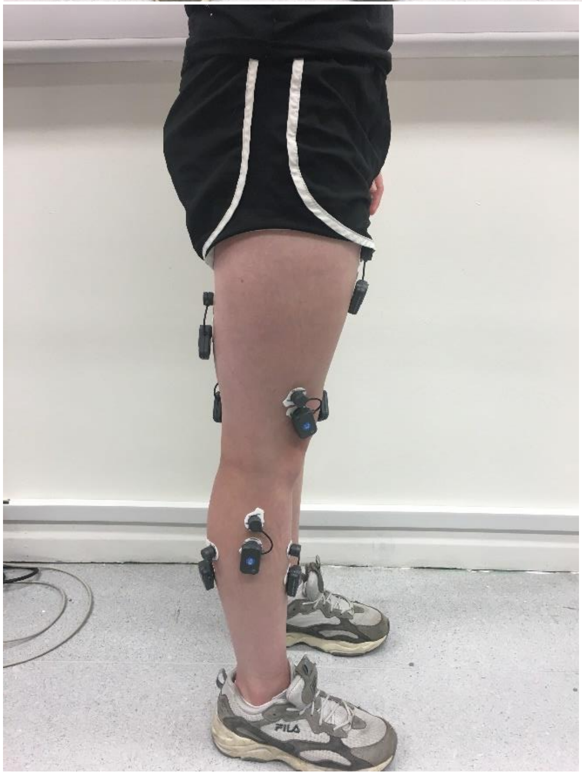
Appendix A

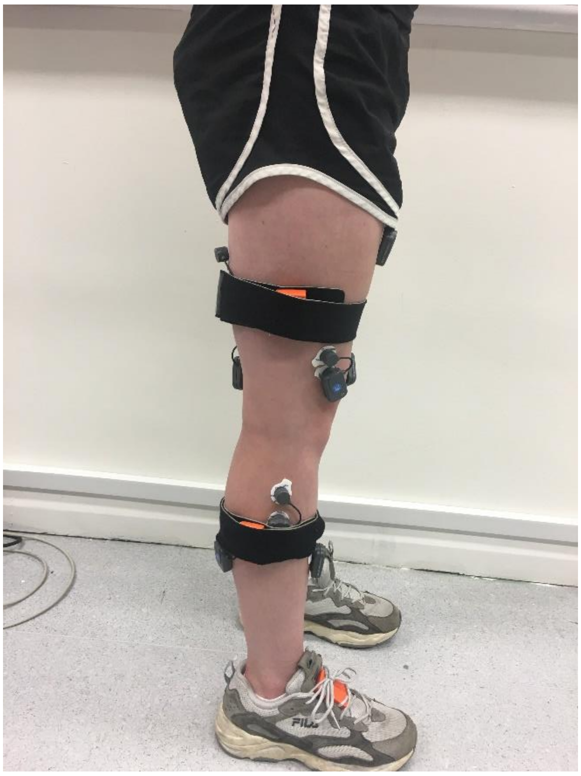
BTS FREEEMG sensor placement





Appendix B

XSENS sensor placement



Appendix C

Examples of participant undergoing exercise trials (walking, steps lateral, steps forward)